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Array substrate and method of manufacturing the same, and display panel

Abstract

The present disclosure provides an array substrate and a method of manufacturing the same and a display panel, which belongs to the field of display technologies. The method of manufacturing the array substrate comprises: providing a base substrate; forming a drive circuit layer on the base substrate, wherein the drive circuit layer includes a switching transistor; forming an insulating material layer on one side of the drive circuit layer distal to the base substrate, wherein the insulating material layer has a connection via-hole exposing at least a part region of a drain electrode of the switching transistor; and forming an electrode layer on one side of the insulating material layer distal to the base substrate, wherein a surface of the electrode layer distal to the base substrate has groove structures extending to the connection via-hole.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of U.S. application Ser. No. 17/297,655 which is the 371 application of PCT Application No. PCT/CN2020/119046, filed on Sep. 29, 2020, the entire contents of both of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) The present disclosure relates to the field of display technology, and more particularly, to an array substrate and a method of manufacturing the same, and a display panel.

BACKGROUND

(2) Thin-film transistor liquid crystal display panels realize the orientation control of liquid crystals through an orientation film, and the quality of the orientation film has a direct impact on the picture effect of the display panel.

(3) In the preparation of the liquid crystal display panels, the orientation film of an array substrate

is prone to poor coating, which reduces the yield of the array substrate and degrades the picture quality of the liquid crystal display panels.

(4) The foregoing information disclosed in this section are provided only for acquiring a better understanding of the background of the present disclosure, and therefore may include information that does not constitute prior art known to those of ordinary skill in the art.

SUMMARY

(5) It is an object of the present disclosure to provide an array substrate and a method of manufacturing the same and a display panel, thus avoiding poor coating of an orientation layer.

(6) In order to achieve the above-mentioned object of the invention, the present disclosure adopts the following technical solutions:

(7) According to a first aspect of the present disclosure, there is provided a method of manufacturing an array substrate, including: providing a base substrate; forming a drive circuit layer on the base substrate, wherein the drive circuit layer includes a switching transistor; forming an insulating material layer on one side of the drive circuit layer distal to the base substrate, wherein the insulating material layer has a connection via-hole exposing at least a part region of a drain electrode of the switching transistor; and forming an electrode layer on one side of the insulating material layer distal to the base substrate, wherein a surface of the electrode layer distal to the base substrate has at least one groove structure extending to the connection via-hole.

(8) In an exemplary embodiment of the present disclosure, forming the electrode layer on one side of the insulating material layer distal to the base substrate includes: sequentially forming a first electrode layer, an insulating dielectric layer, and a second electrode layer which are laminated on one side of the insulating material layer distal to the base substrate, to form the electrode layer.

(9) In an exemplary embodiment of the present disclosure, forming the insulating material layer on one side of the drive circuit layer distal to the base substrate includes: forming an organic insulating layer on one side of the drive circuit layer distal to the base substrate, wherein the organic insulating layer has the connection via-hole exposing at least a part region of the drain electrode of the switching transistor.

(10) In an exemplary embodiment of the present disclosure, the method of manufacturing the array substrate further includes: forming an orientation layer on one side of the electrode layer distal to the base substrate.

(11) In an exemplary embodiment of the present disclosure, the electrode layer is formed such that the second electrode layer includes a plurality of strip sub-electrodes and slits located among the strip sub-electrodes, wherein any one of the groove structures is in communication with one of the slits.

(12) In an exemplary embodiment of the present disclosure, the second electrode layer is formed such that a width of the groove structure is the same as a width of the communicated slit.

(13) In an exemplary embodiment of the present disclosure, the second electrode layer is formed such that an extension direction of the groove structure is the same as an extension direction of the communicated slit.

(14) In an exemplary embodiment of the present disclosure, the second electrode layer is formed such that an orthographic projection of one end of the groove structure close to the slit on the base substrate overlaps with an orthographic projection of one end of the communicated slit close to the groove structure on the base substrate.

(15) In an exemplary embodiment of the present disclosure, forming the electrode layer on one side of the insulating material layer distal to the base substrate includes: forming the first electrode layer on one side of the insulating material layer distal to the base substrate; forming the insulating dielectric layer on one side of the first electrode layer distal to the base substrate; and forming the second electrode layer on one side of the insulating dielectric layer distal to the base substrate, wherein the second electrode layer has drainage grooves, and the drainage grooves extend to the connection via-hole, to serve as the groove structures.

(16) In an exemplary embodiment of the present disclosure, forming the electrode layer on one side of the insulating material layer distal to the base substrate includes: forming the first electrode layer on one side of the insulating material layer distal to the base substrate; forming the insulating dielectric layer on one side of the first electrode layer distal to the base substrate, wherein the insulating dielectric layer has a plurality of drainage grooves extending to the connection via-hole; and forming the second electrode layer on one side of the insulating dielectric layer distal to the base substrate, wherein the second electrode layer covers side walls and a bottom of the drainage groove, to form the groove structure.

(17) In an exemplary embodiment of the present disclosure, forming the electrode layer on one side of the insulating material layer distal to the base substrate includes: forming the first electrode layer on one side of the insulating material layer distal to the base substrate, wherein the first electrode layer has a plurality of drainage grooves, and the drainage grooves extend to the connection via-hole; forming the insulating dielectric layer on one side of the first electrode layer distal to the base substrate; and forming the second electrode layer on one side of the insulating dielectric layer distal to the base substrate, wherein the insulating dielectric layer and the second electrode layer sequentially cover side walls and a bottom of the drainage groove, to form the groove structure.

(18) According to a second aspect of the present disclosure, there is provided an array substrate, including: a base substrate; a drive circuit layer, disposed on one side of the base substrate and including a switching transistor; an insulating material layer, disposed on one side of the drive circuit layer distal to the base substrate, and having a connection via-hole exposing at least a part region of a drain electrode of the switching transistor; and an electrode layer, disposed on one side of the insulating material layer distal to the base substrate, wherein a surface of the electrode layer distal to the base substrate has at least one groove structure extending to the connection via-hole.

(19) In an exemplary embodiment of the present disclosure, the electrode layer includes a first electrode layer, an insulating dielectric layer, and a second electrode layer sequentially laminated on one side of the insulating material layer distal to the base substrate.

(20) In an exemplary embodiment of the present disclosure, the insulating material layer includes an organic insulating layer, wherein the organic insulating layer has the connection via-hole exposing at least a part region of the drain electrode of the switching transistor.

(21) In an exemplary embodiment of the present disclosure, the array substrate further includes: an orientation layer, disposed on one side of the electrode layer distal to the base substrate.

(22) In an exemplary embodiment of the present disclosure, the second electrode layer has a plurality of drainage grooves extending to the connection via-hole, and the drainage grooves serves as the groove structures of the electrode layer.

(23) In an exemplary embodiment of the present disclosure, the insulating dielectric layer has a plurality of drainage grooves extending to the connection via-hole, the second electrode layer is bent inwardly into the drainage groove, to form the groove structure, and an orthographic projection of the groove structure on the insulating dielectric layer is located within the drainage groove.

(24) In an exemplary embodiment of the present disclosure, the first electrode layer has a plurality of drainage grooves extending to the connection via-hole, the insulating dielectric layer and the second electrode layer cover a bottom and side walls of the drainage groove, to form the groove structure, and an orthographic projection of the groove structure on the first electrode layer is located within the drainage groove.

(25) In an exemplary embodiment of the present disclosure, the groove structure has a depth of 300 to 1300 angstroms.

(26) In an exemplary embodiment of the present disclosure, wherein a depth of the groove structure is equal to a thickness of the second electrode layer.

(27) In an exemplary embodiment of the present disclosure, a length of a part of the groove structure within the connection via-hole is not less than 0.5 microns.

- (28) In an exemplary embodiment of the present disclosure, a part of the groove structure within the connection via-hole is located on a side wall of the connection via-hole.
- (29) In an exemplary embodiment of the present disclosure, a length of a part of the groove structure outside the connection via-hole is 2.0~5.0 microns.
- (30) In an exemplary embodiment of the present disclosure, the groove structure has a width of 3.0 to 5.0 microns.
- (31) In an exemplary embodiment of the present disclosure, the second electrode layer includes a plurality of strip sub-electrodes, and slits located among the strip sub-electrodes, wherein any one of the groove structures is in communication with one of the slits.
- (32) In an exemplary embodiment of the present disclosure, a width of the groove structure is the same as a width of the communicated slit.
- (33) In an exemplary embodiment of the present disclosure, an extension direction of the groove structure is the same as an extension direction of the communicated slit.
- (34) In an exemplary embodiment of the present disclosure, an orthographic projection of one end of the groove structure close to the slit on the base substrate overlaps with an orthographic projection of one end of the communicated slit close to the groove structure on the base substrate.
- (35) In an exemplary embodiment of the present disclosure, the second electrode layer includes a plurality of strip sub-electrodes and slits located among the strip sub-electrodes, the strip sub-electrode has a width of 2.5 to 3.5 microns, and the slit has a width of 3.0 to 5.0 microns.
- (36) In an exemplary embodiment of the present disclosure, a ratio of the width of the strip sub-electrode to the width of the slit is 0.5 to 0.8.
- (37) In an exemplary embodiment of the present disclosure, an opening of the connection via-hole is in a rectangular shape, a size of the opening of the connection via-hole close to the base substrate in an extension direction of a scan line in the drive circuit layer is 10-12 microns, and the size of the opening of the connection via-hole close to the base substrate in an extension direction of a data line in the drive circuit layer is 6.5-7.5 microns, and the second electrode layer has a common electrode and covers the connection via-hole, the second electrode layer includes a plurality of strip sub-electrodes, slits located among the strip sub-electrodes, and drainage grooves serving as the groove structures, wherein the strip sub-electrode has a width of 2.5 to 3.5 microns and the slit has a width of 3.0 to 5.0 microns, the drainage grooves are in communication with the slits and extend to the connection via-hole, a length of a part of the drainage groove within the connection via-hole is 6.0 to 6.5 microns, and the drainage groove and the communicated slit have the same width and extension direction.
- (38) In an exemplary embodiment of the present disclosure, the opening of the connection via-hole is in a circular shape, and the opening close to the base substrate has a diameter of 7 to 8 microns, and the second electrode layer has a common electrode and covers the connection via-hole, the second electrode layer includes a plurality of strip sub-electrodes, slits located among the strip sub-electrodes, and drainage grooves serving as the groove structures, wherein the strip sub-electrode has a width of 2.5 to 3.5 microns and the slit has a width of 3.0 to 5.0 microns, the drainage grooves are in communication with the slits and extend to the connection via-hole, a length of a part of the drainage groove within the connection via-hole is 0.8 to 1.2 microns, and the drainage groove and the communicated slit have the same width and extension direction.
- (39) In an exemplary embodiment of the present disclosure, an opening of the connection via-hole is in a rectangular shape, a size of the opening of the connection via-hole close to the base substrate in an extension direction of a scan line in the drive circuit layer is 11.5~13.5 microns, the size of the opening of the connection via-hole close to the base substrate in an extension direction of a data line in the drive circuit layer is 10 to 12 microns, and the size of the connection via-hole in the extension direction of the scan line is greater than the size of the connection via-hole in the extension direction of the data line, and the second electrode layer has a pixel electrode and covers the connection via-hole, the second electrode layer includes a plurality of strip sub-electrodes, slits

located among the strip sub-electrodes, and drainage grooves serving as the groove structures, wherein the strip sub-electrode has a width of 2.5 to 3.5 microns and the slit has a width of 3.0 to 5.0 microns, the drainage grooves are in communication with the slits and extend to the connection via-hole, a length of a part of the drainage groove within the connection via-hole is 1.5 to 2.5 microns, and the drainage groove and the communicated slit have the same width and extension direction.

(40) According to a third aspect of the present disclosure, there is provided a display panel, including the above-described array substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other features and advantages of the present disclosure will become more apparent by detailed description of exemplary embodiments thereof with reference to the accompanying drawings.

(2) FIG. 1 is a schematic flow chart showing a method of manufacturing an array substrate according to an embodiment of the present disclosure.

(3) FIG. 2 is a cross-sectional structural schematic diagram illustrating an array substrate according to an embodiment of the present disclosure.

(4) FIG. 3 is a cross-sectional structural schematic diagram illustrating an array substrate according to an embodiment of the present disclosure.

(5) FIG. 4 is a top view structural schematic diagram of an electrode layer according to an embodiment of the present disclosure.

(6) FIG. 5 is a cross-sectional structural schematic diagram of an electrode layer at PP' of FIG. 4 according to an embodiment of the present disclosure.

(7) FIG. 6 is a cross-sectional structural schematic diagram of an insulating dielectric layer at PP' of FIG. 4 according to an embodiment of the present disclosure.

(8) FIG. 7 is a cross-sectional structural schematic diagram of an electrode layer at PP' of FIG. 4 according to an embodiment of the present disclosure.

(9) FIG. 8 is a cross-sectional structural schematic diagram of a first electrode layer at PP' of FIG. 4 according to an embodiment of the present disclosure.

(10) FIG. 9 is a cross-sectional structural schematic diagram of an electrode layer at PP' of FIG. 4 according to an embodiment of the present disclosure.

(11) FIG. 10 is a schematic diagram illustrating a position relationship between a second electrode layer and a connection via-hole according to an embodiment of the present disclosure.

(12) FIG. 11 is a schematic diagram illustrating a position relationship between a second electrode layer and a connection via-hole according to an embodiment of the present disclosure.

(13) FIG. 12 is a photograph of an array substrate with no groove structure formed thereon after the formation of an orientation layer.

(14) FIG. 13 is a photograph of an array substrate with a groove structure formed thereon after the formation of an orientation layer.

(15) The main components in the figures are denoted by the following reference signs: **100**—base substrate; **200**—drive circuit layer; **201**—semiconductor layer; **202**—gate insulating layer; **203**—gate electrode layer; **204**—interlayer dielectric layer; **205**—source-drain metal layer; **210**—switching transistor; **220**—drain electrode; **300**—insulating material layer; **301**—passivation layer; **302**—organic insulating layer; **310**—connection via-hole; **311**—top opening of connection via-hole; **312**—bottom opening of connection via-hole; **400**—electrode layer; **401**—groove structure; **410**—first electrode layer; **420**—insulating dielectric layer; **430**—second electrode layer; **500**—orientation layer; **611**—strip sub-electrode; **612**—first connection portion; **613**—second connection

portion; **620**—slit; **630**—drainage groove; **710**—data line; **720**—scan line; **730**—pixel electrode; **740**—common electrode.

DETAILED DESCRIPTION

(16) Exemplary embodiments will now be described more fully by reference to the accompanying drawings. However, the exemplary embodiments can be implemented in various forms and should not be understood as being limited to the examples set forth herein; rather, the embodiments are provided so that the present disclosure will be more thorough and complete, and the conception of exemplary embodiments will be fully conveyed to those skilled in the art. The described features, structures or characteristics may be combined in one or more embodiments in any appropriate manner. In the foregoing description, many specific details are provided for fully understanding embodiments of the present disclosure.

(17) In the drawings, the thickness of the regions and layers may be exaggerated for clarity. The same reference numbers will be used throughout the drawings to refer to the same or like structures, and detailed descriptions will be repeated.

(18) The described features, structures or characteristics may be combined in one or more embodiments in any appropriate manner. In the foregoing description, many specific details are provided for fully understanding embodiments of the present disclosure. However, those skilled in the art will recognize that technical solutions of the present disclosure may be practiced without one or more of the specific details, or, other methods, components, materials and so on may be used. In other cases, well-known structures, materials or operations are not shown or described in detail to avoid obscuring the main technical idea of the present disclosure.

(19) Words “including” and “having” are used to mean open-ended inclusion and mean that in addition to the listed elements/components/etc., there may be additional elements/components/etc. Words “first” and “second” are used herein only as markers, and they do not limit the number of objects modified after them.

(20) In the present disclosure, the depth of structure of a groove/hole/slit/etc., refers to the size of structure of the groove/hole/slit/etc., in a direction perpendicular to the base substrate. The thickness of a film layer/structure refers to the size of the film layer/structure in a direction perpendicular to the base substrate. The width of structure of the groove/hole/slit/etc., refers to the size of structure of the slot/hole/slit/etc., on a plane perpendicular to its extension direction and parallel to the base substrate.

(21) In the related technology, when an array substrate for a LCD panel is prepared, poor coating tends to occur when orientation liquid is applied to the array substrate through a transfer print (APR, Asahikaseip Photosensitive Resin) plate by adopting a printing (Roller Coater) manner. This coating defect is especially obvious and serious when the array substrate is provided with an organic insulating layer between an electrode layer and a drive circuit layer.

(22) The inventor has conducted an in-depth study on this problem. The inventor found that an insulating material layer located between the drive circuit layer and the electrode layer needs to be provided with a connection via-hole that exposes at least a part region of a drain electrode; and when the orientation liquid is coated by a printing manner, the orientation liquid tends to accumulate around the connection via-hole instead of entering the connection via-hole, resulting in uneven diffusion of the orientation liquid.

(23) Especially when the insulating material layer includes an organic insulating layer, the organic insulating layer often has a larger thickness, for example, a thickness greater than 2 microns. The connection via-hole has a certain slope of for example 45 to 70°, leading to a larger size of the top of the connection via-hole on the insulating material layer, for example, it may be up to about 10 microns. When the orientation liquid is applied to the surface of the array substrate by a printing manner, due to surface tension, the orientation liquid cannot autonomously diffuse into the connection via-hole having a high-depth and a large aperture, and the orientation liquid builds up and is cured near the connection via-hole, which leads to a relative large height of the orientation

layer at the edge of the connection via-hole, and thus leading to mura of the display panel.

(24) In order to solve the above-mentioned technical problems, the present disclosure provides a method of manufacturing an array substrate. Referring to FIGS. 1 to 4, the method of manufacturing the array substrate includes:

(25) In step S110, a base substrate **100** is provided.

(26) In step S120, a drive circuit layer **200** is formed on the base substrate **100**, wherein the drive circuit layer **200** includes a switching transistor **210**.

(27) In step S130, an insulating material layer **300** is formed on a side of the drive circuit layer **200** distal to the base substrate **100**, wherein the insulating material layer **300** has a connection via-hole **310** exposing at least a part region of a drain (or source) electrode **220** of the switching transistor **210**.

(28) In step S140, an electrode layer **400** is formed on a side of the insulating material layer **300** distal to the base substrate **100**. Referring to FIG. 4, a surface of the electrode layer **400** distal to the base substrate **100** has a groove structure **401** extending to the connection via-hole **310**.

(29) According to the method of manufacturing the array substrate provided by the present disclosure, when the electrode layer **400** is formed, the groove structure **401** that is in communication with the connection via-hole **310** is formed on the surface of the electrode layer **400**. In this way, when an orientation layer **500** is formed, the groove structure **401** may play the role of guiding such that orientation liquid diffuses quickly into the connection via-hole **310** along the groove structure **401**, reducing the difficulty of the orientation liquid to flow into the connection via-hole **310** and increasing the speed of the orientation liquid to flow into the connection via-hole **310**, thus the orientation liquid is evenly diffused and poor coating of the orientation layer **500** is avoided.

(30) According to the method of manufacturing the array substrate provided by the present disclosure, referring to FIGS. 2 to 4, the formed array substrate includes the base substrate **100**, the drive circuit layer **200**, the insulating material layer **300** and the electrode layer **400** laminated in sequence. The drive circuit layer **200** is provided on one side of the base substrate **100** and includes the switching transistor **210**. The insulating material layer **300** is provided on one side of the drive circuit layer **200** distal to the base substrate **100**, and has the connection via-hole **310** that exposes at least a part region of the drain electrode **220** of the switching transistor **210**. The electrode layer **400** is provided on one side of the insulating material layer **300** distal to the base substrate **100**, and the surface of the electrode layer **400** distal to the base substrate **100** has the groove structure **401** extending to the connection via-hole **310**. The array substrate of the present disclosure is manufactured by using the above-described method of manufacturing the array substrate, thus avoiding poor coating of the orientation layer **500**, and improving both the yield of the array substrate and the picture quality of a display panel based on the array substrate.

(31) Hereinafter, the principles, details and effects of the method of manufacturing the array substrate provided by the present disclosure will be further explained and illustrated in conjunction with the accompanying drawings.

(32) In step S110, the base substrate **100** may be provided. The base substrate **100** may be a base substrate **100** of inorganic material or of organic material. For example, in an embodiment of the present disclosure, the material of the base substrate **100** may be a glass material such as soda-lime glass, quartz glass, sapphire glass, etc., or may be a metal material such as stainless steel, aluminum, nickel, etc. In another embodiment of the present disclosure, the material of the base substrate **100** may be Polymethyl methacrylate (PMMA), Polyvinyl alcohol (PVA), Polyvinyl phenol (PVP), Polyether sulfone (PES), polyimide, polyamide, polyacetal, Poly carbonate (PC), Polyethylene terephthalate (PET), Polyethylene naphthalate (PEN) or any combination thereof.

(33) Optionally, before the step S120, a buffer dielectric layer such as a silicon nitride layer, a silicon oxide layer, a silicon oxynitride layer or other inorganic insulating material layers may be formed on the base substrate **100** as well. Then, the drive circuit layer **200** is formed on one side of

the buffer dielectric layer distal to the base substrate **100**.

(34) In step **S120**, the drive circuit layer **200** may be formed on one side of the base substrate **100**. The drive circuit layer **200** may be formed with a plurality of pixel drive circuits, wherein the respective pixel drive circuits are configured to load drive voltages to respective pixel electrodes of the electrode layer **400**. Optionally, referring to FIGS. 2 to 3, any of the pixel drive circuits may include the switching transistor **210**.

(35) In some embodiments, the switching transistor **210** may be a thin film transistor. The thin film transistor may be a top-gate thin film transistor, a bottom-gate thin film transistor, or a dual-gate thin film transistor. The material of an active layer of the thin film transistor may be an amorphous silicon semiconductor material, a low temperature polycrystalline silicon semiconductor material, a metal oxide semiconductor material, an organic semiconductor material, or other types of semiconductor materials. The thin film transistor may be an N-type or P-type thin film transistor. The present disclosure does not limit the specific type of the thin film transistor.

(36) Optionally, referring to FIGS. 2 to 3, the drive circuit layer **200** may include a semiconductor layer **201**, a gate insulating layer **202**, a gate electrode layer **203**, an interlayer dielectric layer **204** and a source-drain metal layer **205** laminated between the base substrate **100** and the insulating material layer **300**. The position relationships of the respective film layers may be determined according to the film layer structure of the thin film transistor. For example, in an embodiment of the present disclosure, the drive circuit layer **200** may include the semiconductor layer **201**, the gate insulating layer **202**, the gate electrode layer **203**, the interlayer dielectric layer **204** and the source-drain metal layer **205** that are sequentially laminated, and the thin film transistor thus formed is the top-gate thin film transistor. For example, in another embodiment of the present disclosure, the drive circuit layer **200** may include the gate electrode layer **203**, the gate insulating layer **202**, the semiconductor layer **201**, the interlayer dielectric layer **204** and the source-drain metal layer **205** laminated in sequence, and the thin film transistor thus formed is the bottom-gate thin film transistor. The drive circuit layer **200** may adopt a double-layer gate structure as well, that is, the gate electrode layer **203** may include a first gate electrode layer and a second gate electrode layer, and the gate insulating layer **202** may include a first gate insulating layer for isolating the semiconductor layer **201** and the first gate electrode layer as well as a second gate insulating layer for isolating the first gate electrode layer and the second gate electrode layer. For example, in an embodiment of the present disclosure, the drive circuit layer **200** may include the semiconductor layer **201**, the first gate insulating layer, the first gate electrode layer, the second gate insulating layer, the second gate electrode layer, the interlayer dielectric layer **204** and the source-drain metal layer **205** laminated in sequence on one side of the base substrate **100**.

(37) Optionally, referring to FIGS. 10 and 11, the drive circuit layer **200** may further include a scan line **720** and a data line **710**. The scan line **720** may be located in the gate electrode layer **203** and be electrically connected to a gate electrode of the switching transistor **210**. The data line **710** may be disposed in the source-drain metal layer **205** and be electrically connected to a source electrode of the switching transistor **210**. A drain electrode **220** of the switching transistor **210** is located in the source-drain metal layer **205** and is electrically connected to a pixel electrode **730** in the electrode layer **400**.

(38) In step **S130**, the insulating material layer **300** may be formed on one side of the drive circuit layer **200** distal to the base substrate **100**, wherein the insulating material layer **300** has the connection via-hole **310** exposing at least a part region of the drain electrode **220** of the switching transistor **210**. In this way, the pixel electrode **730** may be electrically connected to the drain electrode **220** of the switching transistor **210** through the connection via-hole **310**.

(39) Optionally, referring to FIGS. 2 and 3, the insulating material layer **300** may include one or both of a passivation layer **301** and an organic insulating layer **302**.

(40) The passivation layer **301** may be provided on a surface of the source-drain metal layer **205** to protect the source-drain metal layer **205**. The material of the passivation layer **301** may be an

inorganic insulating material such as silicon nitride, silicon oxide or silicon oxynitride, etc.

Exemplarily, in an embodiment of the present disclosure, the material of the passivation layer **301** is silicon nitride.

(41) The material of the organic insulating layer **302** may be an organic material, for example, a resin material. Exemplarily, the material of the organic insulating layer **302** may be polyimide. The organic insulating layer **302** may provide a planarized surface for the electrode layer **400** and balance the stress of the array substrate. In addition, the organic insulating layer **302** may reduce parasitic capacitance between the electrode layer **400** and the source-drain metal layer **205**, thereby reducing the influence of the parasitic capacitance on the charging rate and the aperture ratio.

(42) Optionally, the organic insulating layer **302** may have a thickness greater than 1.5 microns. In this way, the formed connection via-hole **310** may have a depth greater than 1.5 microns. Exemplarily, the organic insulating layer **302** may have a thickness of 2 to 3 microns.

(43) In an embodiment of the present disclosure, in step **S130**, the organic insulating layer **302** may be formed on one side of the drive circuit layer **200** distal to the base substrate **100**, wherein the organic insulating layer **302** has the connection via-hole **310** exposing at least a part region of the drain electrode of the switching transistor. Thus, the formed insulating material layer **300** includes at least the organic insulating layer **302**.

(44) In another embodiment of the present disclosure, in step **S130**, the formed insulating material layer **300** may include the passivation layer **301** and the organic insulating layer **302** laminated sequentially on one side of the source-drain metal layer **205** distal to the base substrate **100**.

(45) Optionally, referring to FIGS. 3-4, the connection via-hole **310** has a bottom opening **312** close to one side of the base substrate **100** and a top opening **311** distal to the one side of the base substrate **100**. Side walls of the connection via-hole **310** are located between the bottom opening **312** and the top opening **311**. Preferably, referring to FIG. 4, a portion of the groove structure **401** in the connection via-hole **310** is located at the side wall of the connection via-hole **310**. In other words, an orthographic projection of the portion of the groove structure **401** in the connection via-hole **310** on the base substrate **100** is located between orthographic projections of the bottom opening **312** and the top opening **311** on the base substrate **100**.

(46) Optionally, the size of the opening of the connection via-hole **310** close to the base substrate **100** may be 6 to 15 microns. In other words, the size of the bottom opening of the connection via-hole **310** may be 6 to 15 microns, to ensure a sufficient contact area between the pixel electrodes **730** of the electrode layer **400** and the drain electrode **220** of the switching transistor **210**, thus reducing the contact resistance.

(47) Optionally, the connection via-hole **310** has a slope of 45° to 75° such that the size of the top opening **311** of the connection via-hole **310** is larger than that of the bottom opening **312** of the connection via-hole **310**. In this way, the connection via-hole **310** may have a proper slope to avoid the break of the electrode layer **400** in the connection via-hole **310** due to a too large slope. The shape of the orthographic projection of the connection via-hole **310** on the base substrate **100** may be a square shape, a rectangular shape, a circular shape, an oval shape or other shapes, and the present disclosure does not impose any limitation on it.

(48) For example, in an embodiment of the present disclosure, the opening of the connection via-hole **310** is in a rectangular shape, the size of the opening of the connection via-hole **310** close to the base substrate **100** in an extension direction of the scan line **720** in the drive circuit layer **200** is 10-12 microns; and the size of the opening of the connection via-hole **310** close to the base substrate **100** in an extension direction of the data line **710** in the drive circuit layer **200** is 6.5-7.5 microns. Further, the organic insulating layer **302** may have a thickness of 2.0 to 2.5 microns. Exemplarily, the organic insulating layer **302** has a thickness of 2.3 microns, the opening of the connection via-hole **310** is in a rectangular shape, the size of the opening of the connection via-hole **310** close to the base substrate **100** in the extension direction of the scan line **720** in the drive circuit layer **200** is 11 microns, and the size of the opening of the connection via-hole **310** close to

the base substrate **100** in the extension direction of the data line **710** in the drive circuit layer **200** is 7 microns.

(49) For another example, in another embodiment of the present disclosure, the opening of the connection via-hole **310** has a circular shape, and the opening close to the base substrate **100** has a diameter of 7 to 8 microns. Further, the organic insulating layer **302** may have a thickness of 2.0 to 2.5 microns. Exemplarily, the organic insulating layer **302** has a thickness of 2.3 microns, the opening of the connection via-hole **310** has a circular shape and the opening close to the base substrate **100** has a diameter of 7.5 microns.

(50) For another example, in another embodiment of the present disclosure, the opening of the connection via-hole **310** is in a rectangular shape, the size of the opening of the connection via-hole **310** close to the base substrate **100** is 11.5 to 13.5 microns in the extension direction of the scan line **720** in the drive circuit layer **200**, the size of the opening of the connection via-hole **310** close to the base substrate **100** is 10 to 12 microns in the extension direction of the data line **710** in the drive circuit layer **200**, and the size of the connection via-hole **310** in the extension direction of the scan line **720** is larger than that of the connection via-hole **310** in the extension direction of the data line **710**. Further, the organic insulating layer **302** may have a thickness of 2.0 to 2.5 microns. Exemplarily, the organic insulating layer **302** has a thickness of 2.3 microns, the opening of the connection via-hole **310** is in a rectangular shape, the size of the opening of the connection via-hole **310** close to the base substrate **100** is 12.5 microns in the extension direction of the scan line **720** in the drive circuit layer **200**, and the size of the opening of the connection via-hole **310** close to the base substrate **100** is 11 microns in the extension direction of the data line **710** in the drive circuit layer **200**.

(51) Optionally, the drain electrode **220** of one switching transistor **210** may have one connection via-hole **310** or a plurality of connection via-holes **310**, as long as the electrical connection of the electrode layer to the drain electrode **220** of the switching transistor **210** is guaranteed. In an embodiment of the present disclosure, the drain electrode **220** of one switching transistor **210** has one connection via-hole **310**, that is, the connection via-hole **310** is provided in one-to-one correspondence with the drain electrode **220** of the switching transistor **210** and the connection via-hole **310** exposes at least a part region of the drain electrode **220** of the corresponding switching transistor **210**.

(52) Optionally, in the step **S140**, the electrode layer **400** may be formed on one side of the insulating material layer **300** distal to the base substrate **100**, wherein the electrode layer **400** includes a first electrode layer **410**, an insulating dielectric layer **420** and a second electrode layer **430** laminated sequentially on the one side of the insulating material layer **300**.

(53) Optionally, the material of the first electrode layer **410** and the second electrode layer **430** is a transparent conductive material, for example, a transparent metal oxide. Exemplarily, the material of both the first electrode layer **410** and the second electrode layer **430** is indium zinc oxide (ITO). Further, optionally, any one of the first electrode layer **410** and the second electrode layer **430** may have a thickness of 300 to 1300 angstroms. Preferably, any one of the first electrode layer **410** and the second electrode layer **430** may have a thickness of 500 to 1000 angstroms. Exemplarily, any one of the first electrode layer **410** and the second electrode layer **430** may have a thickness of 400 to 700 angstroms.

(54) Optionally, the material of the insulating dielectric layer **420** may be an inorganic insulating material such as silicon oxide, silicon nitride, or silicon oxynitride, etc. In an embodiment of the present disclosure, the material of the insulating dielectric layer **420** may be silicon nitride.

(55) Optionally, referring to FIGS. 2 to 3, a common electrode **740** is disposed in one of the first electrode layer **410** and the second electrode layer **430**, and a pixel electrode **730** is disposed in the other one of the first electrode layer **410** and the second electrode layer **430**. In this way, the array substrate is formed with sub-pixels arranged in the form of arrays, and one of the sub-pixels includes the pixel electrode **730**, the common electrode **740** disposed corresponding to the pixel

electrode **730**, and the switching transistor **210** electrically connected to the pixel electrode **730**. The source electrode of the switching transistor **210** is connected to a data line **710** and the gate electrode of the switching transistor **210** is connected to a scan line **720**.

(56) Optionally, the array substrate may further have a via-hole extension region surrounding the top opening of the connection via-hole **310** in the insulating material layer **300**, wherein one or two of the first electrode layer **410**, the insulating dielectric layer **420** and the second electrode layer **430** may not cover the via-hole extension region, such that the electrode layer **400** is recessed in the direction from the surface of the via-hole extension region to the base substrate **100**, to form a ring-shaped recessed region. In this way, the connection via-hole **310** in the present disclosure includes the ring-shaped recessed region, and when the groove structure **401** extends into the ring-shaped recessed region, it means that the groove structure **401** extends into the connection via-hole **310** of the present disclosure.

(57) Exemplarily, in an embodiment of the present disclosure, referring to FIG. 3, the first electrode layer **410** has the common electrode **740**, and the first electrode layer **410** and the insulating dielectric layer **420** expose the drain electrode **220** of the switching transistor **210**. The second electrode layer **430** has the pixel electrode **730**, and the pixel electrode **730** is electrically connected to the drain electrode **220** of the switching transistor **210** through the connection via-hole **310**.

(58) Further, the first electrode layer **410** surrounds the connection via-hole **310** on the insulating material layer **300**, there is a via-hole extension region between the first electrode layer **410** and the top opening of the connection via-hole **310** on the insulating material layer **300**, and no first electrode layer **410** is provided within the via-hole extension region. In this way, when the insulating dielectric layer **420** and the second electrode layer **430** are formed sequentially, the insulating dielectric layer **420** and the second electrode layer **430** are attached on the via-hole extension region along with the shape of the via-hole extension region in sequence, to form the ring-shaped recessed region, and the ring-shaped recessed region is in communication with the connection via-hole **310**. In this way, the connection via-hole **310** of the present disclosure is a region within the ring-shaped recessed region. When the groove structure **401** extends into the ring-shaped recessed region, it means that the groove structure **401** extends into the connection via-hole **310** of the present disclosure.

(59) In another embodiment of the present disclosure, referring to FIG. 2, the first electrode layer **410** has the pixel electrode **730** that is electrically connected to the drain electrode **220** of the switching transistor **210** through the connection via-hole **310**. The insulating dielectric layer **420** covers the first electrode layer **410**, and the second electrode layer **430** has the common electrode **740**. The common electrode **740** may or may not cover the connection via-hole **310**.

(60) Referring to FIG. 4, when the electrode layer **400** is formed, the groove structure **401** extending to the connection via-hole **310** may be disposed on the surface of the electrode layer **400** distal to the base substrate **100**. Thus, the groove structure **401** may guide the orientation liquid to flow into the connection via-hole **310**, avoiding poor coating of the orientation layer **500** due to the accumulation of the orientation liquid at the edge of the connection via-hole **310**. Further, the groove structure **401** extends at least to the edge of the connection via-hole **310**. Preferably, the groove structure **401** extends into the connection via-hole **310**.

(61) Optionally, referring to FIG. 4, a plurality of groove structures **401** extend to the same connection via-hole **310**. In other words, in any pixel region, the electrode layer **400** may have a plurality of groove structures **401**, and the groove structures **401** extend into the same connection via-hole **310**. One pixel region may include a region where a pixel electrode **730** is located and a region where the switching transistor **210** to which the pixel electrode **730** is electrically connected is located.

(62) Optionally, the electrode layer **400** is formed such that the groove structure **401** has a depth of 300 to 1300 angstroms. In this way, the groove structure **401** may have a depth of 300 to 1300 angstroms in the formed array substrate, to improve guiding function that the groove structure **401**

applies on the orientation liquid. Further, the groove structure **401** may have a depth of 500 to 1000 angstroms. Exemplarily, in some embodiments, the groove structure **401** has a depth of 400 to 700 angstroms.

(63) Optionally, the electrode layer **400** is formed such that the groove structure **401** has a width of 3.0 to 5.0 microns. In this way, the situation that the flow-in of the orientation liquid may not be facilitated because the groove structure **401** is too narrow, may be avoided; and the situation that the synergy of the drainage effect of the two edges of the groove structure **401** will not be facilitated because the groove structure **401** is too wide.

(64) Optionally, the electrode layer **400** is formed such that the groove structure **401** has a length not less than 0.5 microns in the connection via-hole **310**. In this way, in the formed array substrate, the groove structure **401** has a length not less than 0.5 microns in the connection via-hole **310**, that is, the groove structure **401** extends into the connection via-hole **310** by not less than 0.5 microns. In this way, the orientation liquid may be more effectively guided to flow into the connection via-hole **310** by the groove structure **401**, thus improving diffusion uniformity of the orientation liquid. In the embodiment of the present disclosure, the length of the groove structure **401** in the connection via-hole **310** may refer to a length of a portion of an orthographic projection of the groove structure **401** on the base substrate **100** that is located within an orthographic projection of the top opening of the connection via-hole **310** on the base substrate **100**. Exemplarily, in an embodiment of the present disclosure, the length of the groove structure **401** in the connection via-hole **310** is 6.25 microns.

(65) Optionally, the electrode layer **400** is formed such that a part of the groove structure **401** outside the connection via-hole **310** has a length of 2.0 to 5.0 microns. In this way, the situation that the part of the groove structure **401** outside the connection via-hole **310** is too short may be avoided, thus ensuring that the groove structure **401** can be in effective communication with the inside and outside of the connection via-hole **310**, so as to ensure that the orientation liquid smoothly flows into the groove structure **401** and then into the connection via-hole **310**. At the same time, squeezing a light-transmitting region of the display panel due to a too large size of the groove structure **401** outside the connection via-hole **310** can be avoided, thus ensuring a larger aperture ratio of the display panel.

(66) Optionally, when the electrode layer **400** is formed, referring to FIGS. **10** and **11**, the second electrode layer **430** may include a plurality of strip sub-electrodes **611** and slits **620** located among the strip sub-electrodes **611**, wherein any of the groove structures **401** is in communication with a slit **620**. In this way, in the formed array substrate, the second electrode layer **430** includes the plurality of strip sub-electrodes **611** and the slits **620** disposed among the strip sub-electrodes **611**, wherein any of the groove structures **401** is in communication with a slit **620**. Optionally, the array substrate may be a single-domain array substrate, where the extension of each of the stripe sub-electrodes **611** of the second electrode layer **430** is a straight line segment; and the array substrate may be a dual-domain array substrate as well, where the extension of each of the stripe sub-electrodes **611** of the second electrode layer **430** is a bent line segment. It will be understood that, when the double-domain structure is used such that the stripe sub-electrodes **611** are bent in the plane where the second electrode layer **430** is located, the slits **620** among the stripe sub-electrodes **611** will be bent accordingly as well. The extension direction of the slit **620** refers to the extension direction of an end of the slit **620**, for example, the extension direction of one end of the slit **620** close to the groove structure **401**.

(67) Optionally, the electrode layer **400** may be formed such that the strip sub-electrode **611** has a width of 2.5 to 3.5 microns, and the slit **620** has a width of 3.0 to 5.0 microns. Further, the ratio of the width of the strip sub-electrode **611** to the width of the slit **620** is 0.5 to 0.8. For example, in an embodiment of the present disclosure, the second electrode layer **430** includes the common electrode **740**, the strip sub-electrode **611** has a width of 2.6 microns, and the slit **620** has a width of 4.8 microns. For further example, in another embodiment of the present disclosure, the second

electrode layer includes the common electrode **740**, the strip sub-electrode **611** has a width of 2.7 microns, and the slit **620** has a width of 3.8 microns. For further example, in another embodiment of the present disclosure, the second electrode layer **430** includes the pixel electrode **730**, the strip sub-electrode **611** has a width of 2.8 microns, and the slit **620** has a width of 4.0 microns.

(68) Since the groove structure **401** is in communication with the slit **620**, the orientation liquid may flow into the connection via-hole **310** along the slit **620** and the groove structure **401** when the orientation layer **500** is formed, which further improves diffusion uniformity of the orientation liquid. In the present disclosure, the communication of the groove structure **401** with the slit **620** means that the orthographic projection of the groove structure **401** on the base substrate **100** at least partially overlaps or contacts with the orthographic projection of the slit **620** on the base substrate **100**.

(69) Further, optionally, the second electrode layer **430** is formed such that the width of the groove structure **401** is the same as that of the slit **620**. In this way, the orientation liquid may more smoothly flows into the connection via-hole **310**.

(70) Further, optionally, the second electrode layer **430** is formed such that the extension direction of the groove structure **401** is the same as that of the slit **620** that the groove structure **401** communicates. In this way, the orientation liquid may more smoothly flow into the connection via-hole **310**.

(71) Further, optionally, the second electrode layer **430** is formed such that an orthographic projection of one end of the groove structure **401** close to the slit **620** on the base substrate **100** overlaps with an orthographic projection of one end of the communicated slit **620** close to the groove structure **401** on the base substrate **100**. In this way, the end of the groove structure **401** is aligned with and adjacent to the end of the communicated slit **620**, such that the orientation liquid may flow into the groove structure **401** from the slit **620** without adjusting its flow direction.

(72) Preferably, the second electrode layer **430** is formed such that the extension direction of the groove structure **401** is the same as the extension direction of the communicated slit **620**, the width of the groove structure **401** is the same as the width of the communicated slit **620**, and the orthographic projection of one end of the groove structure **401** close to the slit **620** on the base substrate **100** overlaps with the orthographic projection of one end of the communicated slit **620** close to the groove structure **401** on the base substrate **100**. In this way, the groove structure **401** is located at the extension position of the slit **620**, which is beneficial to guide the orientation liquid to flow into the connection via-hole **310** along the slit **620** and the groove structure **401**.

(73) When the electrode layer **400** is formed, a drainage groove may be provided on the first electrode layer **410**, the insulating dielectric layer **420** or the second electrode layer **430**, such that the groove structure **401** is formed on the surface of the electrode layer **400**.

(74) In some embodiments of the present disclosure, the electrode layer **400** may be formed by using the following method.

(75) In step **S210**, referring to FIG. 5, the first electrode layer **410** is formed on one side of the insulating material layer **300** distal to the base substrate **100**.

(76) In step **S220**, referring to FIG. 5, the insulating dielectric layer **420** is formed on one side of the first electrode layer **410** distal to the base substrate **100**.

(77) In step **S230**, referring to FIG. 5, the second electrode layer **430** is formed on one side of the insulating dielectric layer **420** distal to the base substrate **100**, wherein the second electrode layer **430** has a plurality of drainage grooves **630** that extend to the connection via-hole **310**, and the drainage grooves **630** form the groove structures **401**.

(78) Thus, in step **S230**, the drainage groove **630** is formed on the second electrode layer **430**, and the drainage groove **630** functions as the groove structure **401** of the electrode layer **400** to guide the orientation liquid to flow into the connection via-hole **310**.

(79) Preferably, the drainage groove **630** penetrates through the second electrode layer **430** in a direction perpendicular to the base substrate **100**. Thus, the depth of the groove structure **401** is

equal to the thickness of the second electrode layer **430**. When the second electrode layer **430** is formed, the drainage groove **630** may be manufactured simultaneously during the patterning of the second electrode layer **430** without adding additional devices and processes.

(80) In the array substrate manufactured according to steps **S210** to **S230**, the second electrode layer **430** has a plurality of drainage grooves **630** extending to the connection via-hole **310**, to enable the electrode layer **400**, to form the groove structure **401**. Preferably, the depth of the groove structure **401** is equal to the thickness of the second electrode layer **430**, that is, the drainage grooves **630** penetrate through the second electrode layer **430** in the direction perpendicular to the base substrate **100**.

(81) Optionally, referring to FIGS. **10** and **11**, the second electrode layer **430** has the plurality of strip sub-electrodes **611** and slits **620** located among the strip sub-electrodes **611**, wherein any of the drainage grooves **630** is in communication with the slit **620**. Wherein, further, the width of the drainage groove **630** is the same as the width of the communicated slit **620**. Further, the extension direction of the drainage groove **630** is the same as the extension direction of the communicated slit **620**. In this way, the array substrate may avoid poor coating of the orientation layer.

(82) In an embodiment of the present disclosure, the drainage groove **630** is in communication with the slit **620**, the extension direction of the drainage groove **630** is the same as the extension direction of the communicated slit **620**, the width of the drainage groove **630** is the same as the width of the communicated slit **620**, the depth of the drainage groove **630** is the same as the depth of the communicated slit **620**, and an orthographic projection of one end of the drainage groove **630** close to the slit **620** on the base substrate **100** overlaps with an orthographic projection of one end of the communicated slit **620** close to the drainage groove **630** on the base substrate **100**. In this way, the drainage groove **630** may be regarded as the extension of the slit **620** into the connection via-hole **310**. In other words, some of the slits **620** of the second electrode layer **430** may extend into the connection via-hole **310**, and the parts of the slits **620** extending into the connection via-hole **310** that exceed other adjacent slits **620** may be regarded as the drainage groove **630** of the present disclosure. This may on the one hand, enable the orientation liquid to smoothly flow into the connection via-hole **310** to overcome poor coating of the orientation layer **500**, on the other hand, ensure the uniformity of the second electrode layer **430** as well, thus ensuring display effect of the display panel.

(83) For verification, in the present disclosure, the array substrate provided with the groove structure and the array substrate provided with no groove structure are manufactured and the orientation layers are formed on the two array substrates by a printing manner. FIG. **12** is a photograph of the array substrate with no groove structure formed thereon after the orientation layer is formed. Referring to FIG. **12**, the array substrate has a large number of bright spots **M**. FIG. **13** is a photograph of the array substrate with the groove structure formed thereon after the orientation layer is formed. Referring to FIG. **13**, the array substrate has good uniformity without the defect such as bright spots, which indicates the uniform diffusion of the orientation liquid.

(84) In an embodiment, the first electrode layer **410** may include the common electrode **740**. Referring to FIG. **10**, the second electrode layer **430** may include the pixel electrode **730**. The pixel electrode **730** may include a plurality of strip sub-electrodes **611** and slits **620** located among the strip sub-electrodes **611**, and may further include a first connection portion **612** and a second connection portion **613**. The first connection portion **612** enables, respective ends of the strip sub-electrodes **611** distal to the switching transistor **210** electrically connected to the pixel electrode **730**, to be electrically connected to one another, and the second connection portion **613** enables, respective ends of the strip sub-electrodes **611** close to the switching transistor **210** electrically connected to the pixel electrode **730**, to be electrically connected to one another. The second connection portion **613** of the pixel electrode **730** is electrically connected to the drain electrode **220** of the switching transistor **210** through the connection via-hole **310**, the second connection portion **613** is provided with the drainage groove **630** serving as the groove structure **401**, and the

drainage groove **630** is in communication with the slit **620** of the pixel electrode **730**.

(85) As an example, the opening of the connection via-hole **310** is in a rectangular shape, the size of the opening of the connection via-hole **310** close to the base substrate **100** is 11.5 to 13.5 microns along the extension direction of the scan line **720** in the drive circuit layer **200**, the size of the opening of the connection via-hole **310** close to the base substrate **100** along the extension direction of the data line **710** in the drive circuit layer **200** is 10 to 12 microns, and the size of the connection via-hole **310** in the extension direction of the scan line **720** is larger than the size of the connection via-hole **310** in the extension direction of the data line **710**. The second electrode layer **430** includes the pixel electrode **730** and covers the connection via-hole **310**, and includes a plurality of strip sub-electrodes **611**, slits **620** located among the strip sub-electrodes **611** and drainage groove **630** serving as the groove structure **401**. The strip sub-electrode **611** has a width of 2.5 to 3.5 microns, and the slit **620** has a width of 3.0 to 5.0 microns. The drainage groove **630** is in communication with the slit **620** and extends into the connection via-hole **310**, and a part of the drainage groove **630** within the connection via-hole **310** has a length of 1.5 to 2.5 microns. The drainage groove **630** is the same as the communicated slit **620** in terms of width and extension direction.

(86) For example, the opening of the connection via-hole **310** is in a rectangular shape, the size of the opening of the connection via-hole **310** close to the base substrate **100** along the extension direction of the scan line **720** in the drive circuit layer **200** is 12.5 microns. The size of the opening of the connection via-hole **310** close to the base substrate **100** is 11 microns in the extension direction of the data line **710** in the drive circuit layer **200**. The second electrode layer **430** includes the pixel electrode **730** and covers the connection via-hole **310**, and the second electrode layer **430** includes a plurality of strip sub-electrodes **611**, slits **620** located among the strip sub-electrodes **611**, and drainage groove **630** serving as the groove structure **401**. The strip sub-electrode **611** has a width of 2.8 microns, and the slit **620** has a width of 4.0 microns. The drainage groove **630** is in communication with the slit **620** and extends into the connection via-hole **310**, and a part of the drainage groove **630** within the connection via-hole **310** has a length of 2 microns. The drainage groove **630** is the same as the communicated slit **620** in terms of width and the extension direction.

(87) Further, the first electrode layer **410** surrounds the connection via-hole **310** on the insulating material layer **300**, and there is a via-hole extension region between the first electrode layer **410** and the top opening of the connection via-hole **310** on the insulating material layer **300**, and no first electrode layer **410** is provided within the via-hole extension region. The insulating dielectric layer **420** and the second electrode layer **430** are attached on the via-hole extension region along with the shape of the via-hole extension region in sequence, to form the ring-shaped recessed region. The connection via-hole **310** of the present disclosure is a region within the ring-shaped recessed region. A part of the groove structure **401** within the connection via-hole **310** is located within the ring-shaped recessed region, and the part of the groove structure **401** within the ring-shaped recessed region has a length of 2 microns.

(88) In an embodiment, the first electrode layer **410** may include the pixel electrode **730**, and the pixel electrode **730** is connected to the drain electrode **220** of the switching transistor **210** through the connection via-hole **310**. The insulating dielectric layer **420** may cover the pixel electrode **730**. Referring to FIG. 11, the second electrode layer **430** may include the common electrode **740**. The common electrode **740** may include a plurality of strip sub-electrodes **611** and slits **620** located among the strip sub-electrodes **611**, and may further include a first connection portion **612** and a second connection portion **613**. The first connection portion **612** enables respective ends of the strip sub-electrodes **611** distal to the switching transistor **210** corresponding to the common electrode **740** to be electrically connected to one another, and the second connection portion **613** enables respective ends of the strip sub-electrodes **611** close to the switching transistor **210** corresponding to the common electrode **740** to be electrically connected to one another. The switching transistor **210** corresponding to the common electrode **740** refers to a switching transistor

210 belonging to the same sub-pixel as the common electrode 740, i.e., a switching transistor 210 electrically connected to the pixel electrode 730 which is located in the same pixel region as the common electrode 740. In the second electrode layer 430, the first connection portions 612 of the common electrodes 740 disposed on the same line may be electrically connected to one another, and the second connection portions 613 of the common electrodes 740 disposed on the same line may be electrically connected to one another. In this way, the uniformity of the common electrodes 740 may be ensured. The second connection portion 613 of the common electrode 740 may cover the connection via-hole 310, the second connection portion 613 is provided with the drainage groove 630 serving as the groove structure 401, and the drainage groove 630 is in communication with the slit 620 of the common electrode 740.

(89) As an example, the opening of the connection via-hole 310 is in a rectangular shape. The size of the opening of the connection via-hole 310 close to the base substrate 100 is 10 to 12 microns in the extension direction of the scan line 720 in the drive circuit layer 200. The size of the opening of the connection via-hole 310 close to the base substrate is 6.5 to 7.5 microns in the extension direction of the data line 710 in the drive circuit layer 200. The second electrode layer 430 includes the common electrode 740 and covers the connection via-hole 310, and the second electrode layer 430 includes a plurality of strip sub-electrodes 611, slits 620 located among the strip sub-electrodes 611, and drainage grooves 630 serving as the groove structures 401. The strip sub-electrode 611 has a width of 2.5 to 3.5 microns, and the slit 620 has a width of 3.0 to 5.0 microns. The drainage groove 630 is in communication with the slit 620 and extends into the connection via-hole 310, and the length of a part of the drainage groove 630 within the connection via-hole 310 is 6.0 to 6.5 microns. The drainage groove 630 and the communicated slit 620 have the same width and extension direction.

(90) For example, the opening of the connection via-hole 310 is in a rectangular shape. The size of the opening of the connection via-hole 310 close to the base substrate 100 is 11 microns in the extension direction of the scan line 720 in the drive circuit layer 200. The size of the opening of the connection via-hole 310 close to the base substrate is 7 microns in the extension direction of the data line 710 in the drive circuit layer 200. The second electrode layer 430 includes the common electrode 740 and covers the connection via-hole 310, and the second electrode layer 430 includes a plurality of strip sub-electrodes 611, slits 620 located among the strip sub-electrodes 611, and drainage grooves 630 serving as the groove structure 401. The strip sub-electrode 611 has a width of 2.6 microns, and the slit 620 has a width of 4.8 microns. The drainage groove 630 is in communication with the slit 620 and extends into the connection via-hole 310, and a part of the drainage groove 630 within the connection via-hole 310 has a length of 6.25 microns. The drainage groove 630 and the communicated slit 620 have the same width and extension direction.

(91) As another example, the opening of the connection via-hole 310 has a circular shape, and the opening close to the base substrate has a diameter of 7 to 8 microns. The second electrode layer 430 has the common electrode 740 and covers the connection via-hole 310, and the second electrode layer 430 includes a plurality of strip sub-electrodes 611, slits 620 located among the strip sub-electrodes 611, and drainage groove 630 serving as the groove structure 401. The strip sub-electrode 611 has a width of 2.5 to 3.5 microns, and the slit 620 has a width of 3.0 to 5.0 microns. The drainage groove 630 is in communication with the slit 620 and extends into the connection via-hole 310, and the length of a part of the drainage groove 630 within the connection via-hole 310 is 0.8 to 1.2 microns. The drainage groove 630 and the communicated slit 620 have the same width and extension direction.

(92) For example, the opening of the connection via-hole 310 has a circular shape, and the opening close to the base substrate has a diameter of 7.5 microns. The second electrode layer 430 has the common electrode 740 and covers the connection via-hole 310, and the second electrode layer 430 includes a plurality of strip sub-electrodes 611, slits 620 located among the strip sub-electrodes 611, and drainage groove 630 serving as the groove structure 401. The strip sub-electrode 611 has a

width of 2.7 microns, and the slit **620** has a width of 3.8 microns. The drainage groove **630** is in communication with the slit **620** and extends into the connection via-hole **310**. A part of the drainage groove **630** within the connection via-hole **310** has a length of 1 micron. The drainage groove **630** and the communicated slit **620** have the same width and extension direction.

(93) In some other embodiments of the present disclosure, the electrode layer **400** may be formed by using the following method.

(94) In step **S310**, referring to FIG. **6**, the first electrode layer **410** is formed on one side of the insulating material layer **300** distal to the base substrate **100**.

(95) In step **S320**, referring to FIG. **6**, the insulating dielectric layer **420** is formed on one side of the first electrode layer **410** distal to the base substrate **100**, wherein the insulating dielectric layer **420** has a plurality of drainage grooves **630** extending to the connection via-hole **310**.

(96) In step **S330**, referring to FIG. **7**, the second electrode layer **430** is formed on one side of the insulating dielectric layer **420** distal to the base substrate **100**, wherein the second electrode layer **430** covers the side wall and bottom of the drainage groove **630**, to form the groove structure **401**.

(97) In this way, when the second electrode layer **430** is formed, the second electrode layer **430** may cover the side wall and bottom of the drainage groove **630** such that the surface of the second electrode layer **430** forms the groove structure **401** in the region where the drainage groove **630** is located. In the formed electrode layer **400**, the insulating dielectric layer **420** has a plurality of drainage slots **630** extending to the connection via-hole **310**, and the second electrode layer **430** is bent inwardly into the drainage grooves **630**, to form the groove structure **401**, wherein an orthographic projection of the groove structure **401** on the insulating dielectric layer **420** is located in the drainage grooves **630**.

(98) Optionally, in step **S320**, the insulating dielectric material layer may be formed on one side of the first electrode layer **410** distal to the base substrate **100** by a deposition manner, and then the insulating dielectric material layer may be patterned to expose the drain electrode **220** of the switching transistor **210** and to form the groove structure **401** that is in communication with the connection via-hole **310**. Exemplarily, a first photoresist layer may be formed on a surface of the insulating dielectric material layer distal to the base substrate **100**, and then the first photoresist layer is exposed and developed by using grayscale exposure techniques, to form a patterned second photoresist layer. The second photoresist layer exposes the insulating dielectric material layer in a first region which covers at least the bottom of the connection via-hole **310**, the thickness of the second photoresist layer in a second region is a first thickness and the second region is a region where the drainage groove **630** is to be formed, the second photoresist layer has a second thickness in a third region, the third region covers the pixel region, and the second thickness is greater than the first thickness. Then, the insulating dielectric material layer is etched in the first region to expose the drain electrode **220** of the switching transistor **210**. Then, the second photoresist layer is ashed such that the insulating dielectric material layer is exposed in the second region and the third region is still covered with photoresist. Then, the insulating dielectric material layer is etched in the second region, to form the drainage groove **630** that is in communication with the connection via-hole **310**. The residual photoresist is removed.

(99) Optionally, in step **S330**, a second electrode material layer may be formed on one side of the insulating dielectric layer **420** distal to the base substrate **100** by a deposition manner, and then the second electrode material layer is patterned, to form the second electrode layer **430**. In this way, planarizing the drainage groove **630** may be avoided, such that the groove structure formed by the drainage groove **630** to be preserved, which in turn allows the groove structure **401** to be formed on the surface of the electrode layer **400** distal to the base substrate **100**.

(100) In some other embodiments of the present disclosure, the electrode layer **400** may be formed by using the following method.

(101) In step **S410**, referring to FIG. **8**, the first electrode layer **410** is formed on one side of the insulating material layer **300** distal to the base substrate **100**, wherein the first electrode layer **410**

has a plurality of drainage grooves **630** which extend to the connection via-hole **310**.

(102) In step **S420**, referring to FIG. **9**, the insulating dielectric layer **420** is formed on one side of the first electrode layer **410** distal to the base substrate **100**.

(103) In step **S430**, referring to FIG. **9**, the second electrode layer **430** is formed on one side of the insulating dielectric layer **420** distal to the base substrate **100**, wherein the insulating dielectric layer **420** and the second electrode layer **430** sequentially cover the side wall and bottom of the drainage groove **630**, to form the groove structure **401**.

(104) In this way, in step **S410**, the first electrode layer **410** may be formed with the drainage groove **630** extending to the connection via-hole **310**, and the recess structure formed by the drainage groove **630** may cause the insulating dielectric layer **420** and the second electrode layer **430** to form a recess, to form the groove structure **401**. In the formed electrode layer **400**, the first electrode layer **410** has a plurality of drainage grooves **630** extending to the connection via-hole **310**, the insulating dielectric layer **420** and the second electrode layer **430** cover the bottom and side walls of the drainage groove **630**, to form the groove structure **401**, and an orthographic projection of the groove structure **401** on the first electrode layer **410** is located in the drainage grooves **630**.

(105) Optionally, in step **S420**, the insulating dielectric material layer may be formed by a deposition manner, and undergo a patterning operation, to form the insulating dielectric layer **420**. In step **S430**, the second electrode material layer may be formed by a deposition manner, and undergo a patterning operation, to form the second electrode layer **430**. Thus, the insulating dielectric layer **420** and the second electrode layer **430** will not planarize the drainage groove **630**, such that the insulating dielectric layer **420** and the second electrode layer **430** form the groove structure **401** along with the shape of the drainage groove **630**.

(106) In other embodiments of the present disclosure, other methods may be used for the formation of the groove structure **401** by the electrode layer **400**, for example, the insulating material layer **300** forms the drainage groove in communication with the connection via-hole **310** such that the electrode layer **400** forms the groove structure **401**. Other feasible methods will not be elaborated in the present disclosure.

(107) Optionally, the method of manufacturing the array substrate of the present disclosure may further include:

(108) In step **S150**, the orientation liquid may be coated on one side of the electrode layer **400** distal to the base substrate **100**, and the orientation layer **500** is then formed. In this step, the orientation liquid can flow into the connection via-hole **310** along the groove structure **401**, to not only fill the connection via-hole **310** but also avoid the accumulation of the orientation liquid near the connection via-hole **310**, which improves the diffusion uniformity of the orientation liquid, then avoids poor coating of the orientation layer **500**.

(109) Optionally, the orientation liquid may be cured by a light curing manner, to form the orientation layer **500**. Exemplarily, an optical alignment device may be used such that the orientation liquid is cured into the orientation layer **500** with a predetermined orientation. The orientation layer **500** may be formed by the orientation liquid by using other manners, for example, formed by using a friction manner after the orientation liquid is cured such that the orientation layer **500** has a predetermined orientation.

(110) The present disclosure further provides an array substrate. Referring to FIGS. **2** to **4**, the array substrate includes: the base substrate **100**; the drive circuit layer **200**, disposed on one side of the base substrate **100**, and including the switching transistor **210**; the insulating material layer **300**, disposed on one side of the drive circuit layer **200** distal to the base substrate **100**, and having the connection via-hole **310** exposing at least a part region of the drain electrode **220** of the switching transistor **210**; and the electrode layer **400**, disposed on one side of the insulating material layer **300** distal to the base substrate **100**, wherein the surface of the electrode layer **400** distal to the base substrate **100** has the groove structure **401** extending to the connection via-hole **310**.

(111) The array substrate provided by the present disclosure may be manufactured by any of the methods of manufacturing the array substrate described in the embodiments of the above-described methods of manufacturing the array substrate, and therefore has the same or similar technical effect, which will not be elaborated herein by the present disclosure.

(112) In an embodiment of the present disclosure, the electrode layer **400** includes the first electrode layer **410**, the insulating dielectric layer **420** and the second electrode layer **430** that are sequentially laminated on one side of the insulating material layer **300** distal to the base substrate **100**.

(113) In an embodiment of the present disclosure, the insulating material layer **300** includes the organic insulating layer **302**, and the organic insulating layer **302** has the connection via-hole **310** that exposes at least a part region of the drain electrode **220** of the switching transistor **210**.

(114) In an embodiment of the present disclosure, the array substrate further includes the orientation layer **500**, and the orientation layer **500** is provided on one side of the electrode layer **400** distal to the base substrate **100**.

(115) In an embodiment of the present disclosure, the groove structure **401** has a depth of 300 to 1300 angstroms.

(116) In an embodiment of the present disclosure, the depth of the groove structure **401** is equal to the thickness of the second electrode layer **430**.

(117) In an embodiment of the present disclosure, the groove structure **401** extends at least to the edge of the connection via-hole **310**.

(118) In an embodiment of the present disclosure, the length of a part of the groove structure **401** with the connection via-hole **310** is not less than 0.5 microns

(119) In an embodiment of the present disclosure, a part of the groove structure **401** with the connection via-hole **310** is located on the sidewall of the connection via-hole **310**.

(120) In an embodiment of the present disclosure, the length of a part of the groove structure **401** outside the connection via-hole **310** is 2.0~5.0 microns.

(121) In an embodiment of the present disclosure, the width of the groove structure **401** is 3.0~5.0 microns.

(122) In an embodiment of the present disclosure, the second electrode layer **430** includes a plurality of strip sub-electrodes **611** and slits **620** located among the strip sub-electrodes **611**, the strip sub-electrode **611** has a width of 2.5~3.5 microns, and the slit **620** has a width of 3.0~5.0 microns.

(123) In an embodiment of the present disclosure, the ratio of the width of the strip sub-electrode **611** to the width of the slit **620** is 0.5 to 0.8.

(124) In an embodiment of the present disclosure, referring to FIGS. **10** and **11**, the second electrode layer **430** includes a plurality of strip sub-electrodes **611** and slits **620** located among the strip sub-electrodes **611**, and any of the groove structures **401** is in communication with one of the slits **620**.

(125) In an embodiment of the present disclosure, referring to FIGS. **10** and **11**, the width of the groove structure **401** is the same as the width of the communicated slit **620**.

(126) In an embodiment of the present disclosure, referring to FIGS. **10** and **11**, the extension direction of the groove structure **401** is the same as that of the communicated slit **620**.

(127) In an embodiment of the present disclosure, referring to FIGS. **10** and **11**, an orthographic projection of one end of the groove structure **401** close to the slit **620** on the base substrate **100**, overlaps an orthographic projection of one end of the communicated slit **620** close to the groove structure **401** on the base substrate **100**.

(128) In an embodiment of the present disclosure, referring to FIG. **5**, the second electrode layer **430** has a plurality of drainage grooves **630** extending to the connection via-hole **310**, and the drainage grooves **630** are used as the groove structures **401** of the electrode layer **400**.

(129) In an embodiment of the present disclosure, referring to FIGS. **6** and **7**, the insulating

dielectric layer **420** has a plurality of drainage grooves **630** extending to the connection via-hole **310**, the second electrode layer **430** is bent inwardly into the drainage groove **630**, to form the groove structure **401**, and an orthographic projection of the groove structure **401** on the insulating dielectric layer **420** is located within the drainage groove **630**.

(130) In an embodiment of the present disclosure, referring to FIGS. **8** and **9**, the first electrode layer **410** has a plurality of drainage grooves **630** extending to the connection via-hole **310**, and the insulating dielectric layer **420** and the second electrode layer **430** cover the bottom and side walls of the drainage groove **630**, to form the groove structure **401**. An orthographic projection of the groove structure **401** on the first electrode layer **410** is located within the drainage groove **630**.

(131) In an embodiment of the present disclosure, the opening of the connection via-hole **310** is in a rectangular shape, the opening of the connection via-hole **310** is in a rectangular shape in the extension direction of the scan line **720** in the drive circuit layer **200**. The size of the opening of the connection via-hole **310** close to the base substrate **100** is 10 to 12 microns in the extension direction of the scan line **720** in the drive circuit layer **200**. The size of the opening of the connection via-hole **310** close to the base substrate is 6.5 to 7.5 microns in the extension direction of the data line **710** in the drive circuit layer **200**. The second electrode layer **430** includes the common electrode **740** and covers the connection via-hole **310**, and the second electrode layer **430** includes a plurality of strip sub-electrodes **611**, slits **620** located among the strip sub-electrodes **611**, and drainage grooves **630** serving as the groove structures **401**. The strip sub-electrode **611** has a width of 2.5 to 3.5 microns, and the slit **620** has a width of 3.0 to 5.0 microns. The drainage groove **630** is in communication with the slit **620** and extends into the connection via-hole **310**, and the length of a part of the drainage groove **630** within the connection via-hole **310** is 6.0 to 6.5 microns. The drainage groove **630** and the communicated slit **620** have the same width and extension direction.

(132) In an embodiment of the present disclosure, the opening of the connection via-hole **310** has a circular shape, and the opening close to the base substrate has a diameter of 7 to 8 microns. The second electrode layer **430** has the common electrode **740** and covers the connection via-hole **310**, and the second electrode layer **430** includes a plurality of strip sub-electrodes **611**, slits **620** located among the strip sub-electrodes **611**, and drainage grooves **630** serving as the groove structures **401**. The strip sub-electrode **611** has a width of 2.5 to 3.5 microns, and the slit **620** has a width of 3.0 to 5.0 microns. The drainage groove **630** is in communication with the slit **620** and extends into the connection via-hole **310**, and the length of a part of the drainage groove **630** within the connection via-hole **310** is 0.8 to 1.2 microns. The drainage groove **630** and the communicated slit **620** have the same width and extension direction.

(133) In an embodiment of the present disclosure, the opening of the connection via-hole **310** is in a rectangular shape, the size of the opening of the connection via-hole **310** close to the base substrate **100** is 11.5 to 13.5 microns in the extension direction of the scan line **720** in the drive circuit layer **200**, the size of the opening of the connection via-hole **310** close to the base substrate **100** is 10 to 12 microns in the extension direction of the data line **710** in the drive circuit layer **200**, and the size of the connection via-hole **310** in the extension direction of the scan line **720** is larger than that of the connection via-hole **310** in the extension direction of the data line **710**. The second electrode layer **430** includes the pixel electrode **730** and covers the connection via-hole **310**, and includes a plurality of strip sub-electrodes **611**, slits **620** located among the strip sub-electrodes **611**, and drainage grooves **630** serving as the groove structures **401**. The strip sub-electrode **611** has a width of 2.5 to 3.5 microns, and the slit **620** has a width of 3.0 to 5.0 microns. The drainage groove **630** is in communication with the slit **620** and extends into the connection via-hole **310**, and the length of a part of the drainage groove **630** within the connection via-hole **310** is 1.5 to 2.5 microns. The drainage groove **630** and the communicated slit **620** have the same width and extension direction.

(134) Further, the first electrode layer **410** surrounds the connection via-hole **310** on the insulating

material layer **300** and there is a via-hole extension region between the first electrode layer **410** and the top opening of the connection via-hole **310** on the insulating material layer **300**, and no first electrode layer **410** is provided within the via-hole extension region. The insulating dielectric layer **420** and the second electrode layer **430** are attached on the via-hole extension region along with the shape of the via-hole extension region in sequence, to form the ring-shaped recessed region. The connection via-hole **310** of the present disclosure is a region within the ring-shaped recessed region. A part of the groove structure **401** within the connection via-hole **310** is located within the ring-shaped recessed region, and the part of the groove structure **401** within the ring-shaped recessed region has a length of 2 microns.

(135) Other details and effects of the array substrate of the present disclosure have been described in detail in the embodiments of the above-described method of manufacturing the array substrate, or can be reasonably deduced from what is recorded in the embodiments of the above-described method of manufacturing the array substrate, which will not be elaborated herein.

(136) Embodiments of the present disclosure further provide a display panel that includes any of the array substrates described in the embodiments of the above-mentioned array substrate. The display panel may be a computer screen, a television screen, a cell phone screen, or other types of liquid crystal display panels. Since the display panel has any of the array substrates described in the embodiments of the above-mentioned array substrate, thus has the same beneficial effect, which will not be elaborated herein.

(137) It shall be understood that the present disclosure does not limit its application to the detailed structures and arrangements of the components presented in this specification. The present disclosure can have other embodiments and can be implemented and carried out in a variety of ways. The foregoing variations and modifications fall within the scope of the present disclosure. It shall be understood that the present disclosure as disclosed and limited by this specification extends to all alternative combinations of two or more individual features mentioned or apparent in the text and/or the accompanying drawings. All such different combinations constitute a plurality of alternative aspects of the present disclosure. Embodiments of this specification illustrate the best mode known for implementing the present disclosure and will enable those skilled in the art to make use of the present disclosure.

Claims

1. An array substrate, comprising: a base substrate; a drive circuit layer, disposed on one side of the base substrate and comprising a switching transistor; an insulating material layer, disposed on one side of the drive circuit layer distal to the base substrate, and having a connection via-hole exposing at least a part region of a drain electrode of the switching transistor; and an electrode layer, disposed on one side of the insulating material layer distal to the base substrate, wherein a surface of the electrode layer distal to the base substrate has at least one groove structure extending to the connection via-hole; wherein the electrode layer comprises a second electrode layer; and an opening of the connection via-hole is in a rectangular shape, a size of the opening of the connection via-hole close to the base substrate in an extension direction of a scan line in the drive circuit layer is 10-12 microns, and the size of the opening of the connection via-hole close to the base substrate in an extension direction of a data line in the drive circuit layer is 6.5-7.5 microns, and the second electrode layer has a common electrode and covers the connection via-hole, the second electrode layer comprises a plurality of strip sub-electrodes, slits located among the strip sub-electrodes, and drainage grooves serving as the groove structures, wherein the strip sub-electrode has a width of 2.5 to 3.5 microns and the slit has a width of 3.0 to 5.0 microns, the drainage grooves are in communication with the slits and extend to the connection via-hole, a length of a part of the drainage groove within the connection via-hole is 6.0 to 6.5 microns, and the drainage groove and the communicated slit have a same width and extension direction.

2. The array substrate according to claim 1, wherein the electrode layer further comprises a first electrode layer and an insulating dielectric layer; and wherein the first electrode layer, the insulating dielectric layer, and the second electrode layer are sequentially laminated on one side of the insulating material layer distal to the base substrate.
3. The array substrate according to claim 1, wherein the insulating material layer comprises an organic insulating layer, and the organic insulating layer has the connection via-hole exposing at least the part region of the drain electrode of the switching transistor.
4. The array substrate according to claim 2, wherein the second electrode layer has a plurality of drainage grooves extending to the connection via-hole, and the drainage grooves serves as the groove structures of the electrode layer; or, the insulating dielectric layer has a plurality of drainage grooves extending to the connection via-hole, the second electrode layer is bent inwardly into the drainage groove, to form the groove structure, and an orthographic projection of the groove structure on the insulating dielectric layer is located within the drainage groove; or, the first electrode layer has a plurality of drainage grooves extending to the connection via-hole, the insulating dielectric layer and the second electrode layer cover a bottom and side walls of the drainage groove, to form the groove structure, and an orthographic projection of the groove structure on the first electrode layer is located within the drainage groove.
5. The array substrate according to claim 1, wherein a depth of the groove structure is equal to a thickness of the second electrode layer.
6. The array substrate according to claim 1, wherein a part of the groove structure within the connection via-hole is located on a side wall of the connection via-hole.
7. A display panel, comprising the array substrate of claim 1.
8. An array substrate, comprising: a base substrate; a drive circuit layer, disposed on one side of the base substrate and comprising a switching transistor; an insulating material layer, disposed on one side of the drive circuit layer distal to the base substrate, and having a connection via-hole exposing at least a part region of a drain electrode of the switching transistor; and an electrode layer, disposed on one side of the insulating material layer distal to the base substrate, wherein a surface of the electrode layer distal to the base substrate has at least one groove structure extending to the connection via-hole; wherein the electrode layer comprises a second electrode layer; and an opening of the connection via-hole is in a circular shape and the opening close to the base substrate has a diameter of 7 to 8 microns, and the second electrode layer has a common electrode and covers the connection via-hole, the second electrode layer comprises a plurality of strip sub-electrodes, slits located among the strip sub-electrodes, and drainage grooves serving as the groove structures, wherein the strip sub-electrode has a width of 2.5 to 3.5 microns and the slit has a width of 3.0 to 5.0 microns, the drainage grooves are in communication with the slits and extend to the connection via-hole, a length of a part of the drainage groove within the connection via-hole is 0.8 to 1.2 microns, and the drainage groove and the communicated slit have a same width and extension direction.
9. The array substrate according to claim 8, wherein the electrode layer further comprises a first electrode layer and an insulating dielectric layer; and wherein the first electrode layer, the insulating dielectric layer, and the second electrode layer are sequentially laminated on one side of the insulating material layer distal to the base substrate.
10. The array substrate according to claim 8, wherein the insulating material layer comprises an organic insulating layer, and the organic insulating layer has the connection via-hole exposing at least the part region of the drain electrode of the switching transistor.
11. The array substrate according to claim 9, wherein the second electrode layer has a plurality of drainage grooves extending to the connection via-hole, and the drainage grooves serves as the groove structures of the electrode layer; or, the insulating dielectric layer has a plurality of drainage grooves extending to the connection via-hole, the second electrode layer is bent inwardly into the drainage groove, to form the groove structure, and an orthographic projection of the groove

structure on the insulating dielectric layer is located within the drainage groove; or, the first electrode layer has a plurality of drainage grooves extending to the connection via-hole, the insulating dielectric layer and the second electrode layer cover a bottom and side walls of the drainage groove, to form the groove structure, and an orthographic projection of the groove structure on the first electrode layer is located within the drainage groove.

12. The array substrate according to claim 8, wherein a depth of the groove structure is equal to a thickness of the second electrode layer.

13. The array substrate according to claim 8, wherein a part of the groove structure within the connection via-hole is located on a side wall of the connection via-hole.

14. An array substrate, comprising: a base substrate; a drive circuit layer, disposed on one side of the base substrate and comprising a switching transistor; an insulating material layer, disposed on one side of the drive circuit layer distal to the base substrate, and having a connection via-hole exposing at least a part region of a drain electrode of the switching transistor; and an electrode layer, disposed on one side of the insulating material layer distal to the base substrate, wherein a surface of the electrode layer distal to the base substrate has at least one groove structure extending to the connection via-hole; wherein the electrode layer comprises a second electrode layer; and an opening of the connection via-hole is in a rectangular shape, a size of the opening of the connection via-hole close to the base substrate in an extension direction of a scan line in the drive circuit layer is 11.5~13.5 microns, the size of the opening of the connection via-hole close to the base substrate in an extension direction of a data line in the drive circuit layer is 10 to 12 microns, and the size of the connection via-hole in the extension direction of the scan line is greater than the size of the connection via-hole in the extension direction of the data line, and the second electrode layer has a pixel electrode and covers the connection via-hole, the second electrode layer comprises a plurality of strip sub-electrodes, slits located among the strip sub-electrodes, and drainage grooves serving as the groove structures, wherein the strip sub-electrode has a width of 2.5 to 3.5 microns and the slit has a width of 3.0 to 5.0 microns, the drainage grooves are in communication with the slits and extend to the connection via-hole, a length of a part of the drainage groove within the connection via-hole is 1.5 to 2.5 microns, and the drainage groove and the communicated slit have a same width and extension direction.

15. The array substrate according to claim 14, wherein the electrode layer further comprises a first electrode layer and an insulating dielectric layer; and wherein the first electrode layer, the insulating dielectric layer, and the second electrode layer are sequentially laminated on one side of the insulating material layer distal to the base substrate.

16. The array substrate according to claim 14, wherein the insulating material layer comprises an organic insulating layer, and the organic insulating layer has the connection via-hole exposing at least the part region of the drain electrode of the switching transistor.

17. The array substrate according to claim 15, wherein the second electrode layer has a plurality of drainage grooves extending to the connection via-hole, and the drainage grooves serves as the groove structures of the electrode layer; or, the insulating dielectric layer has a plurality of drainage grooves extending to the connection via-hole, the second electrode layer is bent inwardly into the drainage groove, to form the groove structure, and an orthographic projection of the groove structure on the insulating dielectric layer is located within the drainage groove; or, the first electrode layer has a plurality of drainage grooves extending to the connection via-hole, the insulating dielectric layer and the second electrode layer cover a bottom and side walls of the drainage groove, to form the groove structure, and an orthographic projection of the groove structure on the first electrode layer is located within the drainage groove.

18. The array substrate according to claim 14, wherein a depth of the groove structure is equal to a thickness of the second electrode layer.

19. The array substrate according to claim 14, wherein a part of the groove structure within the connection via-hole is located on a side wall of the connection via-hole.

